

Electrical, Excavation & Radiography Permits

Locking out electrical energy, disturbing the ground, and a planned radiography permit

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Flutter / Dart

Electrical Isolation

LOTO

Excavation

Radiography
(placeholder)

1. Electrical Isolation Permit

Covers any job that needs electrical equipment to be **de-energised and physically locked and tagged**. The permit is cyan, tagged “LOTO / Electrical Safety”, and is the most lock-out-focused of the six.

A different Section 3

Instead of a gas log, Section 3 is an **Isolation Verification Register**. For each isolation point it records the point number, the equipment ID, the isolation method, the lock tag number, who isolated it and the date and time. Work only starts once every point is on the list.

- **Authorisation:** issuer, receiver (the electrician), an authorising engineer, and a sign-off for closure and **re-energisation**.
- **Six general checklist groups:** permit and job control, electrical safety, energy control, work environment, tools and equipment, and worker competency and PPE.
- **General hazard groups:** electrical hazards (shock, arc flash, short circuit, induced voltage, stored energy), isolation hazards (incomplete lock-out, wrong equipment identified, back-feed) and equipment and process hazards.

Activities and what makes each one different

Activity	Main hazards	Extra checklist groups
Equipment Isolation	Mechanical movement on start-up, incorrect identification, back-feed	Equipment isolation safety, start-up prevention
Panel / Switchgear Isolation	Busbar exposure, capacitor energy, high fault current	Panel safety, stored energy discharge

Activity	Main hazards	Extra checklist groups
Cable / Circuit Isolation	Wrong circuit, hidden live cable, insulation failure	Cable isolation, circuit tracing safety
Lockout / Tagout	Unauthorised energising, missing isolation point, human error	LOTO compliance, key control
Testing for Dead	Residual charge, wrong probe contact, induced voltage	Verification safety, probe technique
Stored Energy Discharge	Capacitor flash, spring release, hydraulic surge	Stored energy control, discharge sequence
Temporary Power Isolation	Temporary re-energising, feedback from a temporary supply	Temporary circuit safety
Maintenance on Isolated Equipment	Residual voltage, unexpected restart, internal live parts	Maintenance safety on isolated equipment
Energisation / Reinstatement	Missed pre-energising check, live-work injury, surge	Reinstatement checklist, energisation sequence

2. Excavation Permit

Covers **ground-disturbing work**: hand digging, machine excavation, trenches, pavement cutting and backfilling. The permit is purple and carries the tags “Ground Risk”, “Utility Hazard” and “Cave-in Risk”.

- **Section 1** records the excavation location, its **dimensions** (length, width, depth), the **ground type** and whether the **utility survey** has been confirmed.
- **Authorisation**: issuer, receiver and a site engineer for mechanical or deep work.
- **Nine general checklist groups**: permit and job control, pre-excavation safety, work-area safety, excavation condition, equipment safety, protection measures, atmosphere and exposure, emergency readiness and simultaneous work control.
- **General hazard groups**: ground and collapse, energy (buried cables, gas lines and water mains), equipment and process, work environment and human factors.
- **Gas testing** is shown for trench and pit work when an atmosphere hazard is flagged.

Activities and what makes each one different

Activity	Main hazards	Extra checklist groups
Manual Excavation	Cave-in, buried-utility strike, overburden fall	Hand tool safety, soil stability, utility proximity
Mechanical Excavation	Bucket strike on utilities, machine tip-over, pedestrian crush	Plant safety, exclusion zone, operator competency

Activity	Main hazards	Extra checklist groups
Trench / Pit / Foundation	Wall collapse, oxygen deficiency, groundwater	Shoring or box, wall stability, depth monitoring, atmosphere
Near Underground Utilities	Gas, electric or water rupture	Utility survey verification, permit-to-dig, hand-dig zone
Road / Pavement Cutting	Traffic, dust and silica, reinstatement failure	Traffic management, dust suppression, road closure
Backfilling / Reinstatement	Poor compaction, disturbing buried services, subsidence	Backfill method approval, layer compaction, record update

3. Radiography Permit — Placeholder

A Radiography permit screen exists, titled “Radiography Permit — Select NDT Method”, in pink. It shows three tiles, each opening a “coming soon” popup:

Tile	Description on the tile
X-Ray Radiography	Industrial X-ray, weld inspection, NDT
Gamma Ray Radiography	Ir-192, Se-75 and Co-60 radioactive sources
Radiation Exclusion Zone	Controlled zone, supervised zone, barriers

Not yet connected

The permit hub currently shows six permit cards. The Radiography screen is built as a placeholder but is not yet linked from the hub, so its full form is still to come.